Introduction

Classical mechanics is intuitive; things move in predictable ways. An experienced ballplayer can take a quick look at a fly ball, and from its location and its velocity, know where to run in order to be there just in time to catch the ball. Of course a sudden unexpected gust of wind might fool him, but that's only because he didn't take into account all the variables. There is an obvious reason why classical mechanics is intuitive: humans, and animals before them, have been using it many times every day for survival. But no one ever used quantum mechanics before the twentieth century. Quantum mechanics describes things so small that they are completely beyond the range of the human senses. So it stands to reason that we did not evolve an intuition for the quantum world. The only way we can comprehend it is by rewiring our intuitions with abstract mathematics. Fortunately, for some odd reason, we did evolve the capacity for such rewiring.

Ordinarily, we learn classical mechanics first, before even attempting quantum mechanics. But quantum physics is much more fundamental than classical physics. As far as we know, quantum mechanics provides an exact description of every physical system, but some things are massive enough that quantum mechanics can be reliably approximated by classical mechanics. That's all that classical mechanics is: an approximation. From a logical point of view, we should learn quantum mechanics first, but very few physics teachers would recommend that. Even this course of lectures—the Theoretical Minimum series—began with classical mechanics. Nevertheless, in these quantum lectures, classical mechanics will play almost no role except near the end, well after the basic principles of quantum mechanics have been explained. I think this is really the right way